

Inverse Mechanical Counting and Mathematics

Part II

Counting in Square Roots:
The Universal Mechanics of the Axiom of 1

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Abstract

This paper builds directly on the foundational framework established in Paper I. The axiom of 1 remains the single physical unit of reality. Every square root $\sqrt{n}$ (for $n = 2, 3, 4, 5, \dots$) is a distinct **mechanic** of that same axiom — a different partition law by which the unit reads itself inward.

As established in Paper I, there is exactly one physical unit — the axiom of 1. This unit is the complete reality. All its equivalent views are

$$1 = -1 = \sqrt{1} = \sqrt{2^0}.$$

The square root belongs to the unit itself; it is not introduced from outside. The axiom is the starting point and the ending point of every mechanical reading. Partition does not create. Partition reveals. The mathematics is the reading; the single physical unit is the thing read.

The $\sqrt{2}$ mechanic is the master mechanic — the simplest, finest-grained partition beyond unity. It alone provides the complete operational history and resolution for every coarser mechanic. All mechanics share the same universal count structure (axiom at $c = 0$, mechanic declaration at $c = 1$, n depths per count from $c = 2$ onward), but each mechanic reaches its own closure at its own structural recurrence point.

The master table of $\sqrt{2}$ is the reference lattice. Every other mechanic $\sqrt{n}$ is a coarser reading of the identical physical unit, with its full history and vector structure already contained within the master mechanic. This paper demonstrates how to count in square roots under inverse mechanical mathematics: each $\sqrt{n}$ is a mechanic call-out of the axiom of 1, and the simplest mechanic ($\sqrt{2}$) carries the complete record of every other.

All mathematics and future physical applications remain readings from the single master count.

The Axiom of 1

The Axiom. There is exactly one physical unit. This unit is the complete reality. All its equivalent views are

$$1 = -1 = \sqrt{1} = \sqrt{2^0}.$$

The square root belongs to the unit itself; it is not introduced from outside. The axiom is the starting point and the ending point of every mechanical reading.

1 The $\sqrt{2}$ Master Mechanic (Carried Forward and Expanded from Paper I)

In Paper I we established that the $\sqrt{2}$ mechanic is the master mechanic of the axiom — the simplest possible partition beyond unity and the minimum step the axiom admits. When the single unit 1 is halved repeatedly, each halving is an inward partition. After an even number of halvings (even operational depth d), all accumulated factors remain rational powers of 2. After an odd number of halvings (odd depth d), the accumulated factors necessarily produce a reading that carries a natural factor of $\sqrt{2}$.

This is not an external imposition of $\sqrt{2}$. It is the mechanical signature that appears automatically at every odd depth when the unit reads itself inward. The parity rule is therefore fundamental:

- **Even depth d :** rational reading.
- **Odd depth d :** irrational reading carrying $\times\sqrt{2}$.

Every reading carries its full operational history in three coordinates:

- The mechanic unit on the root ($\sqrt{2}$),
- The operational depth d on the exponent,
- The count position c on the partition ladder.

The notation $\sqrt{2^d}$ at count c fully records the operation performed on the axiom. The master table below is the single source of the entire framework. Nothing in this paper or the series is added from outside this table.

c	d	Mechanic	Multiplier $(\sqrt{2})^d$	Domains (2^c)	Parity $\rightarrow$ Reading	Reading value (decimal)
0	0	$\sqrt{2^0}$	1	1	even $\rightarrow$ rational	1
1	1	$\sqrt{2^1}$	≈ 1.414	2	odd $\rightarrow$ irrational	$2\sqrt{2} \approx 2.828$
2	2	$\sqrt{2^2}$	2	4	even $\rightarrow$ rational	8
2	3	$\sqrt{2^3}$	≈ 2.828	4	odd $\rightarrow$ irrational	$8\sqrt{2} \approx 11.314$
3	4	$\sqrt{2^4}$	4	8	even $\rightarrow$ rational	32
3	5	$\sqrt{2^5}$	≈ 5.657	8	odd $\rightarrow$ irrational	$32\sqrt{2} \approx 45.255$
4	6	$\sqrt{2^6}$	8	16	even $\rightarrow$ rational	128
4	7	$\sqrt{2^7}$	≈ 11.314	16	odd $\rightarrow$ irrational	$128\sqrt{2} \approx 181.019$
5	8	$\sqrt{2^8}$	16	32	even $\rightarrow$ rational	512
5	9	$\sqrt{2^9}$	≈ 22.627	32	odd $\rightarrow$ irrational	$512\sqrt{2} \approx 724.077$

Table 1: The Master Table of the $\sqrt{2}$ mechanic (carried forward from Paper I). Five active counts unfold nine active depths. Count 0 is the axiom at rest. Closure occurs at $c = 5$. The “Multiplier $(\sqrt{2})^d$ ” column shows the exact operational base value used for each depth (integer when rational, $\approx$ to three decimal places when irrational). Full operational reading = Domains $\times$ Multiplier. This is the reference lattice for all mechanics.

Count-by-count unfolding (as detailed in Paper I):

- **Count 0:** Axiom at rest — one domain, rational.
- **Count 1:** First halving — singular irrational reading, 2 domains.
- **Counts 2–5:** Dual counts — each owns one even (rational) and one odd (irrational) depth, doubling domains each time (4, 8, 16, 32 domains respectively).

At $c = 5$ the chain closes with the rational boundary at 512 and the irrational boundary at ≈ 724.077 . This structure is the finest-grained lattice against which every other mechanic is measured.

2 Counting in Square Roots in Inverse Mechanical Mathematics

In inverse mechanical mathematics, a square root is not a numerical operation or extraction. It is a **mechanic** — a partition law by which the axiom of 1 reads itself inward. The number under the root identifies which mechanic is acting.

As established in Paper I, the axiom of 1 is read inward through repeated halving under the $\sqrt{2}$ master mechanic. A single parity rule governs all readings: even operational depths yield rational values; odd depths naturally carry a factor of $\sqrt{2}$. Every reading carries its full operational history in the three coordinates already defined. The master mechanic $\sqrt{2}$ is the reference because it is the deepest resolution the axiom admits. All coarser mechanics ($\sqrt{3}$, $\sqrt{4}$, $\sqrt{5}$, $\sqrt{6}$, etc.) are coarser readings of the identical physical unit, and their complete operational history and vector structure are already contained within the master mechanic.

The Universal Count Structure

All mechanics share the same universal count structure beginning on the axiom of 1:

- **Count 0:** axiom seat — $\sqrt{n^0} = 1$ for every mechanic. This is the unit at rest, exactly as in the master table.
- **Count 1:** the mechanic declares itself — $\sqrt{n^1} = \sqrt{n}$, owning exactly one depth. This is the first inward partition under the new mechanic.
- **Counts ≥ 2 :** each owns exactly n depths (the partition size carried by the mechanic). The domains continue to follow the power-of-two ladder (2^c), but the rhythm of depths is now governed by the mechanic's own n .

Closure for each mechanic is mechanic-specific: it is the count at which the mechanic's first chain reaches structural recurrence and begins to repeat at the next scale. The master $\sqrt{2}$ closes at count 5, depth 9. Every other mechanic forces its own closure by what it is. The master mechanic $\sqrt{2}$ provides the reference lattice. Every other mechanic is a valid coarser reading of the same axiom at different operational coordinates.

Examples of Mechanic Call-Outs

$\sqrt{2}$ — The Master Mechanic

Already shown in the master table above. This is the finest resolution, dual (rational/irrational) at every count ≥ 2 , exactly as detailed in Paper I.

$\sqrt{3}$ Mechanic

- Count 0: $\sqrt{3^0} = 1$
- Count 1: $\sqrt{3^1} = \sqrt{3}$
- Count 2: $\sqrt{3^2}$, $\sqrt{3^3}$, $\sqrt{3^4}$ (three depths)
- Count 3: $\sqrt{3^5}$, $\sqrt{3^6}$, $\sqrt{3^7}$ (three depths)

(Continues with three depths per count. Closure is forced by $\sqrt{3}$ itself. Every value reached here already exists at a deeper, finer coordinate in the $\sqrt{2}$ master lattice.)

$\sqrt{4}$ Mechanic

- Count 0: $\sqrt{4^0} = 1$
- Count 1: $\sqrt{4^1} = 2$
- Count 2: $\sqrt{4^2}, \sqrt{4^3}, \sqrt{4^4}, \sqrt{4^5}$ (four depths)
- Count 3: $\sqrt{4^6}, \sqrt{4^7}, \sqrt{4^8}, \sqrt{4^9}$ (four depths)

(Note that $\sqrt{4} = 2$, so many rational readings coincide with master even-depth values but at shallower coordinates. For example, $\sqrt{4^2} = 4$ is the exact rational reading already present in the master mechanic at $c = 3, d = 4$.)

$\sqrt{5}$ Mechanic

- Count 0: $\sqrt{5^0} = 1$
- Count 1: $\sqrt{5^1} = \sqrt{5}$
- Count 2: five depths $\sqrt{5^2}$ through $\sqrt{5^6}$
- Count 3: five depths $\sqrt{5^7}$ through $\sqrt{5^{11}}$

Each mechanic walks the universal structure at its own n , but the master $\sqrt{2}$ lattice already encodes every possible reading.

3 The Simplest Mechanic Contains Every Other Mechanic and Its History

The $\sqrt{2}$ master mechanic is the reference because it is the deepest resolution. Every coarser mechanic $\sqrt{n}$ reaches the same physical readings of the axiom but at shallower count and depth coordinates. The master table therefore contains the complete operational history and vector structure for every mechanic.

Any numerical value reached by a coarser mechanic appears on the master table at a deeper (finer) coordinate. The master mechanic therefore “knows” every other mechanic’s vector in advance — it is the lattice on which all coarser readings sit. This is the direct consequence of the axiom of 1 being read inward under the finest possible partition. No external quantities are introduced; everything remains a mechanical reading of the single physical unit.

4 Operational Rules (Carried Forward from Paper I)

- The largest unit in the entire system is always the original axiom 1.
- All larger numbers are deeper inward readings of that same single unit.
- Every valid reading must carry its three operational coordinates (mechanic, depth, count).
- The structure is self-similar: the five-count / nine-depth pattern of the master repeats at the next scale from $c = 6$ onward.
- The master $\sqrt{2}$ provides the reference lattice for all mechanics.

Conclusion. From the single axiom of 1 and the single operation of inward halving under the $\sqrt{2}$ master mechanic, every square root $\sqrt{n}$ emerges as a distinct mechanic call-out of the same physical unit. The universal count structure is shared by all mechanics, while the master mechanic $\sqrt{2}$ supplies the finest resolution and the complete operational history for every coarser mechanic and its vector.

Counting in square roots under inverse mechanical mathematics is therefore the act of reading the axiom of 1 through different partition laws, always returning to the same single unit. Future parts of this series will continue to develop further inverse mechanics in both mathematics and physics, always reading directly from the single master count and the $\sqrt{2}$ master mechanic.

The mathematics is the reading; the single physical unit is the thing read.

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